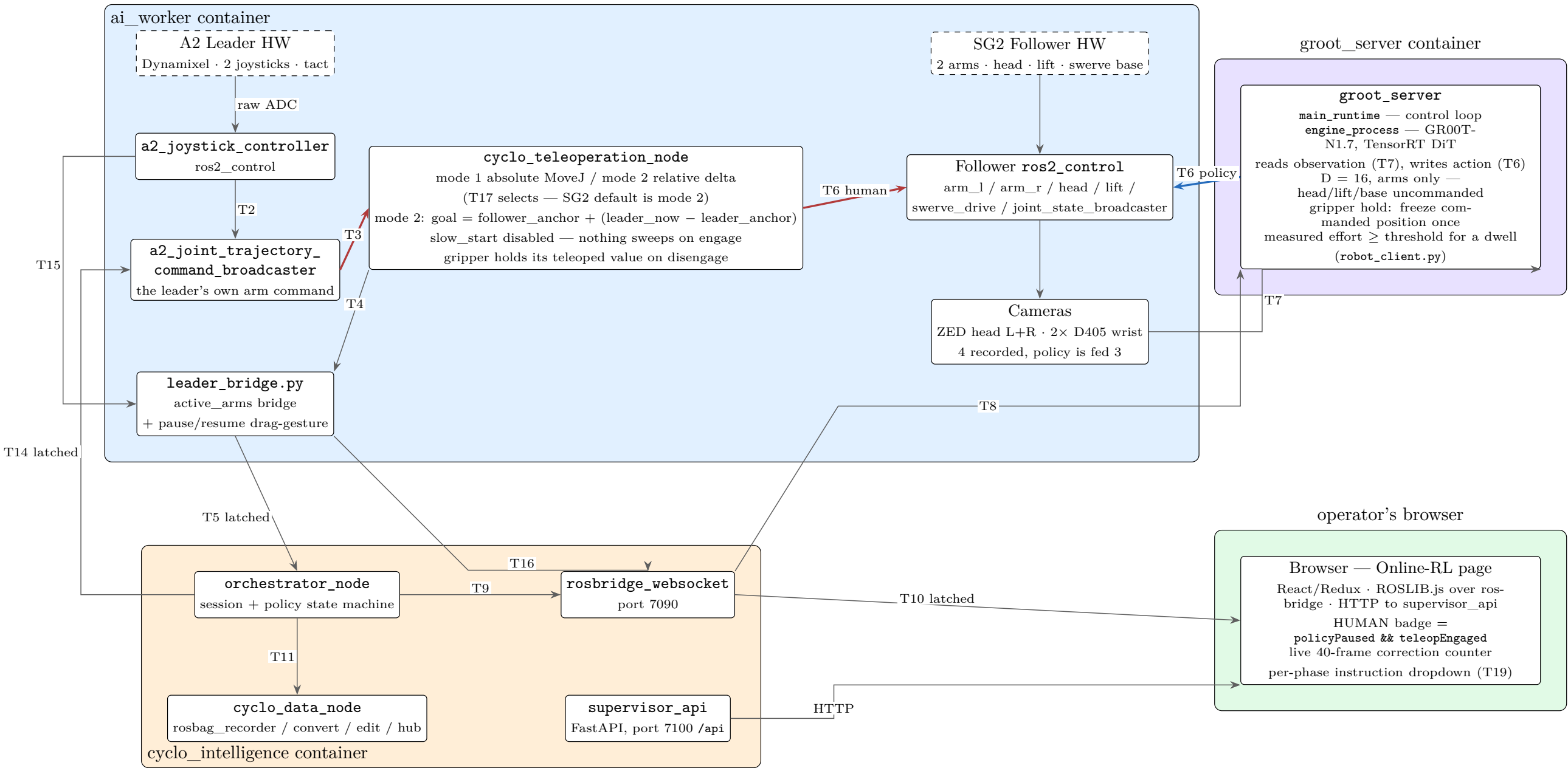


HG-Dagger / Online-RL — Runtime Control Path

FFW SG2 follower · A2 leader · GR00T-N1.7 policy

— human commands the arms — policy commands the arms — status / feedback / control plane. Edge labels are IDs defined in the legend on p.3.



The operator's takeover loop

- joystick drag-down $\geq 1s$ → T16 → INFERENCING → PAUSED
- tact press → T2 COMMAND_TOGGLE → that arm engages
- tact press again → arm disengages, gripper holds its teleoped value
- joystick drag-down $\geq 1s$ → T16 → PAUSED → INFERENCING

The gaps 1→2 and 3→4 are PAUSED with no arm engaged: labelled -1, not human.

T16 is one topic with two meanings, resolved by the phase it arrives in.

T6 is ONE topic, written by BOTH sides
`/leader/joint_trajectory_command_broadcaster_{left,right}/joint_trajectory`

Teleop node and policy publish to the same place, so *action* is valid in either mode — no stale-leader pose. T14 stops them contending.

T18 Home goes straight from the browser to *ai_worker* over rosbridge — the only UI action that skips the orchestrator.

Recording and Conversion — what survives to training

Every recorded topic, and where the online-RL labels are lost

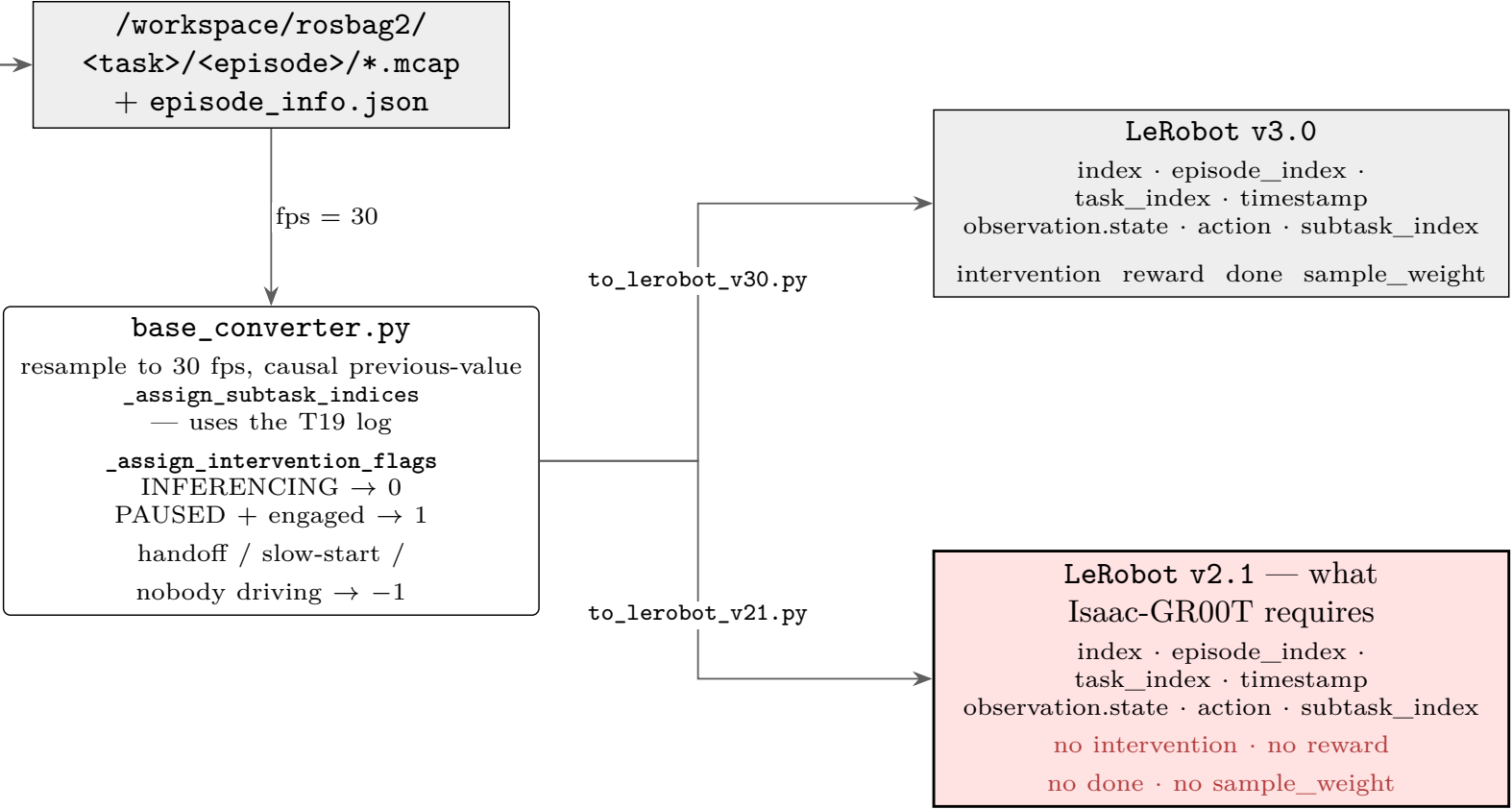
recorded topics — `ffw_sg2_rev1_config.yaml`

observation.images — 4 cameras
cam_left_head /zed/zed_node/left/image_rect_color/compressed
cam_left_wrist /camera_left/.../image_rect_raw/compressed
rot 270
cam_right_wrist /camera_right/.../image_rect_raw/compressed
rot 270
cam_right_head
/zed/zed_node/right/image_rect_color/compressed

observation.state — 22 dims
upper_body /joint_states JointState — 19 joints:
arm_l_joint1..7, gripper_l_joint1,
arm_r_joint1..7, gripper_r_joint1,
head_joint1..2, lift_joint
mobile /odom Odometry — linear_x, linear_y, angular_z

action — 20 dims. Each topic is BOTH the policy’s
command
target and the record target.
arm_left
/leader/joint_trajectory_command_broadcaster_left/...
arm_right
/leader/joint_trajectory_command_broadcaster_right/...
head /leader/joystick_controller_left/joint_trajectory
lift /leader/joystick_controller_right/joint_trajectory
mobile /cmd_vel Twist

recording.extra_topics — in the bag, not training data
/task/inference_status ← the intervention source
/tf
/zed/zed_node/{left,right}/camera_info
/camera_{left,right}/.../camera_info
/leader/teleoperation/control_status deliberately absent —
robotis_interfaces is not installed where the recorder runs.



This is where HG-Dagger dies. `_assign_intervention_flags` computes a correct per-frame label, and `to_learnerobot_v30.py` writes it out (lines 1484–1489). `to_learnerobot_v21.py` never does — `grep -c intervention` on it returns 0. Isaac requires v2.1, so every GR00T export silently loses the human/policy label and degrades to plain behaviour cloning. Fix: *IMPROVEMENTS.md* §3.

Topic / Service Legend

Verified against cyclo_intelligence_backup@main and aiw_backup@feature-a2, 2026-08-30.

ID	Kind	Name	Type	Meaning / gotcha
T1	PUB/SUB	/leader/joystick_controller/tact_trigger	std_msgs/String	Tap emits "left"/"right". Record trigger disabled — one press also engages that arm, which during INFERENCING would put the leader onto T6 against the policy.
T2	PUB/SUB	.../joystick_command	robotis_interfaces/ TeleoperationCommand	ENABLE / DISABLE / TOGGLE, per arm. The tact sends TOGGLE — engage and stop are the same press.
T3	PUB/SUB	.../raw_joint_trajectory	trajectory_msgs/ JointTrajectory	The leader’s own arm command, before mode handling.
T4	PUB/SUB	/leader/teleoperation/control_status	robotis_interfaces/ ControlModeStatus	active_arms + mode. Use T5.
T5	PUB/SUB latched	/leader/teleoperation/active_arms_str	std_msgs/String	Same information as T4 as a plain string, from leader_bridge.py. The UI badge reads it; the bag should too.
T6	PUB/SUB	/leader/joint_trajectory_command_ broadcaster_{left,right}/joint_trajectory	trajectory_msgs/ JointTrajectory	Written by BOTH the teleop node and the policy. That is why action is valid in either mode. T14 prevents contention, not separate topics.
T7	PUB/SUB	/joint_states, /odom, 4 camera topics	JointState, Odometry, CompressedImage	Follower feedback and vision. 4 cameras recorded, 3 declared to the policy.
T8	PUB/SUB	/<backend>/engine_command	std_msgs/String	Orchestrator → groot_server: load / unload / run.
T9	SERVICE	/task/command	SendCommand.srv	Start / Pause / Resume / Clear policy; Start / Stop / Cancel recording.
T10	PUB/SUB latched	/task/inference_status	interfaces/InferenceStatus	inference_phase: READY 0, LOADING 1, INFERENCING 2, PAUSED 3. Transitions only — the converter forward-fills. Sole source of the intervention label.
T11	SERVICE	/data/recording	RecordingCommand.srv	Recording lifecycle, orchestrator → cyclo_data_node.
T13	SERVICE	/data/convert, /data/convert/status	StartConversion.srv, GetConversionStatus.srv	LeRobot conversion. Both convert_v21 and convert_v30 false means "run both".
T14	PUB/SUB latched	/task/policy_active	std_msgs/Bool	Mirrors T10’s INFERENCING into ai_worker. Gates the leader chain off T6 so the two never publish together.
T15	PUB/SUB	.../sensorxel_joy_values	std_msgs/Float64MultiArray	Raw joystick axes. The gesture detector watches these.
T16	PUB/SUB	/leader/teleoperation/toggle_pause_request	std_msgs/Empty	Fires once on a ≥1s drag-hold. One topic, two meanings: pause if INFERENCING, resume if PAUSED. Lets the operator work without the browser.
T17	PUB/SUB	.../control_command	robotis_interfaces/ ControlModeCommand	Mode 1 absolute MoveJ, mode 2 relative delta. SG2 default is mode 2.
T18	PUB/SUB latched	/task/home_in_progress	std_msgs/Bool	One-shot Home trajectory, published straight from the browser — the only UI action that skips the orchestrator.
T19	SERVICE	/task/command (UPDATE_INSTRUCTION)	SendCommand.srv	Appends to instruction_change_log, which splits the episode into correctly-timed sub-segments. Without it a continuous take gets one segment and every frame carries the last instruction set.

Derived dataset columns

- **observation.state** (22) and **action** (20) — they differ: state carries head_joint1/2, action does not. Slice order follows YAML insertion order. The policy declares only the first 16 (both arms); the remainder are recorded but not trained on.
- **task_index** / **subtask_index** — per frame, from episode_info.json’s segments, split by instruction_change_log (T19). The episode’s top-level task_instruction is a separate stable field; it is not the live per-phase instruction, which is overwritten on every T19 click.
- **intervention** — 1 = human drove, 0 = policy drove, −1 = excluded (handoff, nobody driving, phase never observed — note the handoff rule assumes a mode-1 sweep, see IMPROVEMENTS.md Defect 2). This is the inverse of task_is_policy, the name some downstream tooling expects; getting the polarity wrong raises nothing and trains on exactly the wrong frames. v3.0 only.
- **reward**, **done**, **sample_weight** — emitted alongside intervention. v3.0 only, so HIL-SERL-style training is also impossible from a v2.1 export.